

---

```

% Inclass 3a
clear; clc; close all;

f = @(x) x.^2/10-2*sin(x);

x = linspace(-10,20,400);
f_1 = f(x);
plot(x,f_1,'LineWidth',2)
hold on
grid on
axis([-1 2 -1 5])
xlabel('x', 'FontSize', 14)
ylabel('f(x)', 'FontSize', 14);

fprintf('Iteration      x1      x2      x3      x4 \n');
% Parabolic Interpolation
x1 = 0; x3 = 1.5; x2=0.8;
f1 = f(x1); f2 = f(x2); f3 = f(x3);
plot(x1,f1,'o',x2,f2,'o',x3,f3,'o', 'MarkerFaceColor','g')

y1 = Parabolic_I(f, x1,x2,x3,x);
plot(x,y1,'--')

xopt = x2;
for i=1:30
    iter = i;
    C1 = (x2-x1)^2*(f2-f3)-(x2-x3)^2*(f2-f1);
    C2 = (x2-x1)*(f2-f3)-(x2-x3)*(f2-f1);
    x4 = x2 -C1/2/C2;
    f4 = f(x4);
    plot(x4,f4,'o', 'MarkerFaceColor','r')
    pause
    fprintf('%6d %9.4f %9.4f %9.4f %9.4f\n',i,x1,x2,x3,x4);

    x3 = x2; x2 = x1; x1 = x4;
    f3 = f2; f2 = f1; f1 = f4;

    y1 = Parabolic_I(f, x1,x2,x3,x);
    plot(x,y1,'--')
    pause
    if x4 ~= 0, ea = abs((x4 - xopt)/x4)*100; end
    if ea <= 1e-4 || iter > 50, break,
end
xopt = x4;
end
hold on

function [y] = Parabolic_I(f, xa,xb,xc,x)

fa = f(xa);
fb = f(xb);
fc = f(xc);

```

---

---

```

y = fa*(x-xb).*(x-xc)./(xa-xb)./(xa-xc) ...
+ fb*(x-xa).*(x-xc)./(xb-xa)./(xb-xc) ...
+ fc*(x-xa).*(x-xb)./(xc-xa)./(xc-xb);
end

fprintf('\nMinimum point found:\n');
fprintf('x_min = %.6f\n',x4);
fprintf('f(x_min) = %.6f\n',f4);

plot(x4,f4,'ms','MarkerSize',10,'MarkerFaceColor','m');
hold off

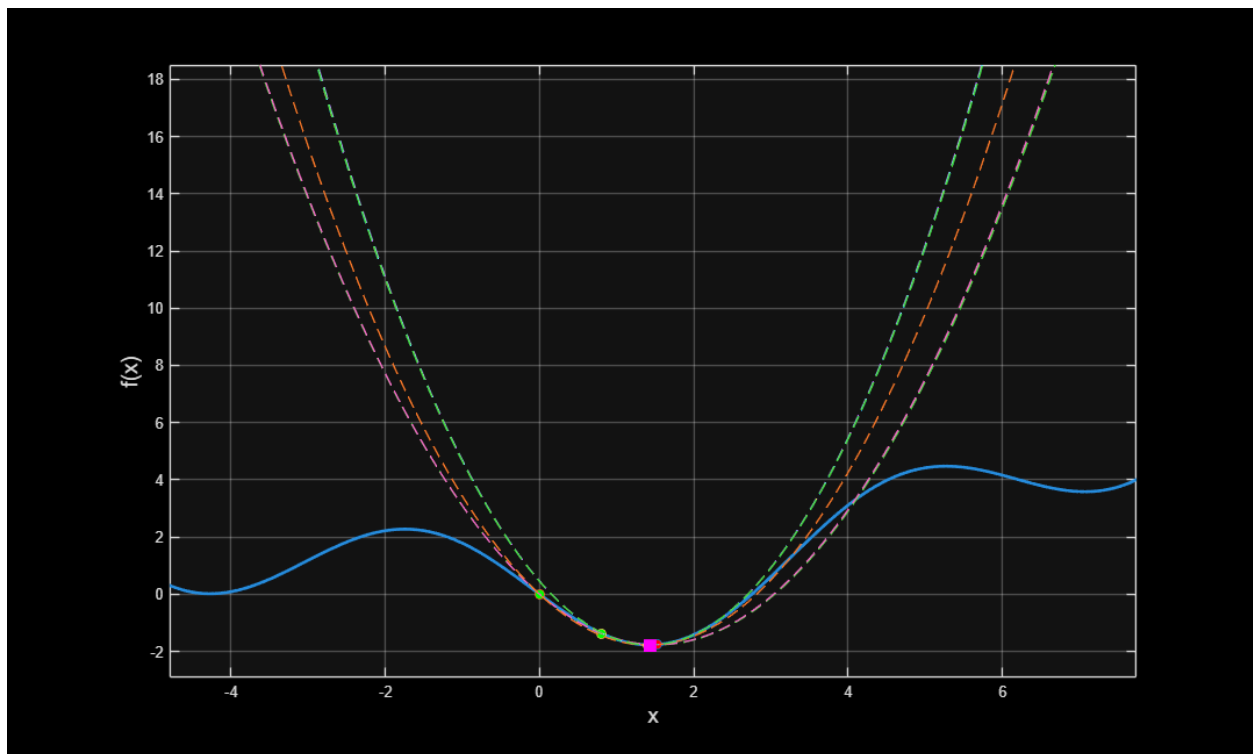
```

| Iteration | x1     | x2     | x3     | x4     |
|-----------|--------|--------|--------|--------|
| 1         | 0.0000 | 0.8000 | 1.5000 | 1.5243 |
| 2         | 1.5243 | 0.0000 | 0.8000 | 1.5177 |
| 3         | 1.5177 | 1.5243 | 0.0000 | 1.4071 |
| 4         | 1.4071 | 1.5177 | 1.5243 | 1.4276 |
| 5         | 1.4276 | 1.4071 | 1.5177 | 1.4275 |
| 6         | 1.4275 | 1.4276 | 1.4071 | 1.4276 |
| 7         | 1.4276 | 1.4275 | 1.4276 | 1.4276 |

```

Minimum point found:
x_min = 1.427552
f(x_min) = -1.775726

```



Published with MATLAB® R2025a